

CSDS HW (Week 8)

Question 1) We will use the `interactive_regression()` function from `CompStatsLib` again.

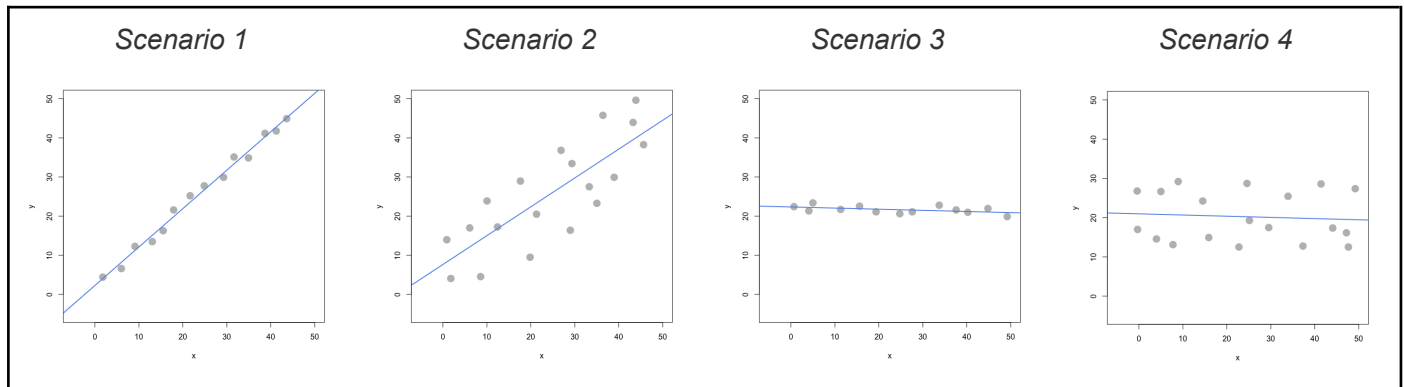
To answer the questions below, understand each of these four scenarios by simulating them:

Scenario 1: Consider a very narrowly dispersed set of points that have a negative or positive steep slope

Scenario 2: Consider a widely dispersed set of points that have a negative or positive steep slope

Scenario 3: Consider a very narrowly dispersed set of points that have a negative or positive shallow slope

Scenario 4: Consider a widely dispersed set of points that have a negative or positive shallow slope



- Comparing scenarios 1 and 2, which do we expect to have a stronger R^2 ?
- Comparing scenarios 3 and 4, which do we expect to have a stronger R^2 ?
- Comparing scenarios 1 and 2, which do we expect has bigger/smaller SSE, SSR, and SST? (intuitively)
- Comparing scenarios 3 and 4, which do we expect has bigger/smaller SSE, SSR, and SST? (intuitively)

Question 2) Let's analyze the `programmer_salaries.txt` dataset we saw in class. Read the file using `read.csv("programmer_salaries.txt", sep="\t")` because the columns are separated by tabs (`\t`).

- Use the `lm()` function to estimate the regression model `Salary ~ Experience + Score + Degree`
Show the beta coefficients, R^2 , and the first 5 values of $\hat{y}$ (`$fitted.values`) and ϵ (`$residuals`)
- Use only linear algebra and the geometric view of regression to estimate the regression yourself:
 - Create an `X` matrix that has a first column of 1s followed by columns of the independent variables (*only show the code*)
 - Create a `y` vector with the Salary values (*only show the code*)
 - Compute the `beta_hat` vector of estimated regression coefficients (*show the code and values*)
 - Using the above, compute a `y_hat` vector of estimated $\hat{y}$ values, and a `res` vector of residuals (*show the code and the first 5 values of `y_hat` and `res`*)
 - Using only the results from (i) – (iv), compute SSR, SSE and SST (*show the code and values*)
- Compute R^2 for in two ways, and confirm you get the same results (*show code and values*):
 - Use any combination of SSR, SSE, and SST
 - Use the squared correlation of vectors `y` and `y_hat`

(see question 3 on next page)

Question 3) We're going to look back at the early days of global car manufacturing, when American, Japanese, and European cars competed to rule the world. Look at the data in file `auto-data.txt`. We are interested in explaining what kind of cars have higher fuel efficiency (i.e., higher *mpg*).

1. *mpg*: miles-per-gallon (dependent variable)
2. *cylinders*: cylinders in engine
3. *displacement*: size of engine
4. *horsepower*: power of engine
5. *weight*: weight of car
6. *acceleration*: acceleration ability of car
7. *model_year*: year model was released
8. *origin*: place car was designed (1: USA, 2: Europe, 3: Japan)
9. *car_name*: make and model names

Note that the data has missing values ('?' in data set), and lacks a header row with variable names:

```
auto <- read.table("auto-data.txt", header=FALSE, na.strings = "?")
names(auto) <- c("mpg", "cylinders", "displacement", "horsepower", "weight",
               "acceleration", "model_year", "origin", "car_name")
```

- a. Let's first try exploring this data and problem:
 - i. Visualize the data as you wish to answer the following questions (report only relevant plots)
 - ii. Report a correlation table of all variables, rounding to two decimal places
(in the `cor()` function, set `use="pairwise.complete.obs"` to handle missing values)
 - iii. From the visualizations and correlations, which variables appear to relate to *mpg*?
 - iv. Which relationships might not be linear? (*don't worry about linearity for rest of this HW*)
 - v. Are there any pairs of independent variables that are highly correlated ($r > 0.7$)?
- b. Let's create a linear regression model where *mpg* is dependent upon all other suitable variables
(*Note: origin is categorical with three levels, so use `factor(origin)` in `lm(...)` to split it into two dummy variables*)
 - i. Which independent variables have a 'significant' relationship with *mpg* at 1% significance?
 - ii. Looking at the coefficients, is it possible to determine which independent variables are the *most effective* at increasing *mpg*? If so, which ones, and if not, why not? (hint: units!)
- c. Let's try to resolve some of the issues with our regression model above.
 - i. Create *fully standardized* regression results: are these slopes easier to compare?
(note: consider if you should standardize *origin*)
 - ii. Regress *mpg* over each *nonsignificant* independent variable, individually.
Which ones become significant when we regress *mpg* over them individually?
 - iii. Plot the distribution of the *residuals*: are they normally distributed and centered around zero?
(get the residuals of a fitted linear model, e.g. `regr <- lm(...)`, using `regr$residuals`)